$= zx' + xy'$ (Postulate 2(b))
$= x'z + xy'$ (Postulate 3(b))

4. $xy + x'z + yz = (xy + x'z) + yz \cdot 1$ (Postulate 2(b))
$= (xy + x'z) + yz \cdot (x + x')$ (Postulate 5(a))
$= (xy + x'z) + (yz \cdot x + yz \cdot x')$ (Postulate 4(a))
$= (xy + x'z) + (x \cdot yz + x' \cdot yz)$ (Postulate 3(b))
$= (xy + x'z) + (xy \cdot z + x' \cdot y \cdot z)$ (Theorem 4(b))
$= ((xy + x'z) + xy \cdot z) + (x' \cdot yz)$ (Theorem 4(a))
$= (xy \cdot z + (xy + x'z)) + (x' \cdot yz)$ (Postulate 3(a))
$= ((xy \cdot z + xy) + x'z) + (x' \cdot yz)$ (Theorem 4(a))
$= ((xy \cdot z + xy \cdot 1) + x'z) + (x' \cdot yz)$ (Postulate 2(b))
$= (xy \cdot (z + 1) + x'z) + (x' \cdot yz)$ (Postulate 4(a))
$= (xy \cdot 1 + x'z) + (x' \cdot yz)$ (Theorem 2(a))
$= (xy + x'z) + (x' \cdot yz)$ (Postulate 2(b))
$= xy + (x'z + x' \cdot yz)$ (Theorem 4(a))
$= xy + (x' \cdot yz + x'z)$ (Postulate 3(a))
$= xy + (x' \cdot zy + x'z)$ (Postulate 3(b))
$= xy + ((x'z) \cdot y + x'z)$ (Theorem 4(b))
$= xy + ((x'z) \cdot y + (x'z) \cdot 1)$ (Postulate 2(b))
$= xy + ((x'z) \cdot (y + 1))$ (Postulate 4(a))
$= xy + ((x'z) \cdot 1)$ (Theorem 2(a))
$= xy + x'z$ (Postulate 2(b))

5. $(x + y)(x' + z)(y + z) = (x + y)(x' + z)$ by duality from function 4

2.4.2 Complement of a Function

- The complement of a function F is F' and is obtained from an interchange of 0's for 1's and 1's for 0's in the value of F.
- De Morgan's theorems can be extended to three or more variables.

$(A + B + C)' = (A + X)'$ (let $B + C = X$)
$= A'X'$ (by Theorem 5(a)) $= A' \cdot (B + C)'$ (substitute $B + C = X$)
$= A' \cdot (B' \cdot C')$ (by Theorem 5(a))
$= A' \cdot B' \cdot C'$ (by Theorem 4(b))